Modelo Competencia con información asimétrica :

We consider the forgetting model with parameter $\rho \in (0,1)$ and states $x_t = (x_{B,t}, x_{A,t})$ that represents the successes of sellers A and B at time t. For this model we consider that each seller $j \in \{A,B\}$ observes only its outcomes x_j. Also each of them has two products $i \in \{1,2\}$. The choice of the client is denoted by y_t and the outcome at time t is denoted by u_t.

Timing: ① customer chooses provider j *

② Provider chosses product i

③ outcome and update state

↳ $\mathbb{P}(u=1) = p_i$

For this model we use $\rho \in (\frac{1}{2}, 1)$ so we don't have to deal with the cantor set and we assume we are in the border ? $x \in [0,1]^2$

* For this model we consider that $y_t \in \{A,B\}$ and

$$\mathbb{P}(y_t = A \mid x_t) = q_j(x_t) = \Lambda(x_{j,t} - x_{-j,t})$$

↳ continuous, increasing

$\Lambda(-z) = 1 - \Lambda(z) \Rightarrow q_1 + q_2 = 1$

TRANSITIONS : $T^s(x) = \rho x + (1-\rho)$, $T^f(x) = \rho x$

Fix the strategies $\sigma_j : K_\rho \to \underbrace{\Delta(\{1,2\})}_{\text{conjunto de distribuciones de prob sobre los prod.}}$ and write $\bar{p}_j = \sum_i \sigma_j(i \mid x_j)\, p_i$

These probabilities is like the average success probability given the strategy σ_j.

From state x we have four possible transitions

$(T^s(x_A), x_B)$ $\quad (T^f(x_A), x_B)$ $\quad (x_A, T^s(x_B))$ $\quad (x_A, T^f(x_B))$

Can we calculate the probability of each transition?

A selected, success $\Rightarrow$ $q_A(x) \cdot \bar{p}_A(x_A)$

A selected, fail $\Rightarrow$ $q_A(x)\,(1-\bar{p}_A(x_A))$

B selected, success $\Rightarrow$ $(1-q_A(x))\,\bar{p}_B(x_B)$

B selected, fail $\Rightarrow$ $(1-q_A(x))\,(1-\bar{p}_B(x_B))$

TRASITION KERNEL (simil de transition matrix para estados continuos)

$$P^\sigma(x,\cdot) = q_A(x)\left[\bar{p}_A(x_A)\,\delta_{(T^s(x_A),x_B)} + (1-\bar{p}_A(x_A))\,\delta_{(T^f(x_A),x_B)}\right]$$

$$+ (1-q_A(x))\left[\bar{p}_B(x_B)\,\delta_{(x_A,T^s(x_B))} + (1-\bar{p}_B(x_B))\,\delta_{(x_A,T^f(x_B))}\right]$$

→ Delta Dirac $\quad \delta(a,b) = \begin{cases} 1 & (a,b)\in E \\ 0 & (a,b)\notin E \end{cases}$

Let $\pi^\sigma(\cdot) = \int_{\mathcal{X}^2} P^\sigma(x,\cdot)\,\pi^\sigma(dx)$ we assume that it exists and that is unique. (why?)

Consider $\eta_j^\sigma(x) := \mathbb{E}^{\pi^\sigma}[q_j(x_t) \mid x_{j,t} = x] = \int \Lambda(x-x')\,\pi^\sigma(dx' \mid x_j = x)$

↳ this is the expected demand that only depends on x_j not on x_{-j} because it is averaging over the state distribution of $-j$.

Under this demand, our model becomes

$$P^{\pi^\sigma}(x_{j,t+1} \in \cdot \mid x_{j,t} = x_A) = \eta_A(x_A)\left[\bar{p}_A(x_A)\,\delta_{T^s(x_A)} + (1-\bar{p}_A(x_A))\,\delta_{T^f(x_A)}\right]$$

$$+ (1-\eta_A(x_A))\,\delta_{x\in\cdot}$$

Now this only depends on the info from provider j.